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B A N G A L O R E • I N D I A

Mapping Agricultural Landscapes: A Country-Wise Analysis of Production, Productivity, and Area

BY

**Babitha Ravindra(2441616)
2 BCA 'B'**

Assignment Submitted to:

Ms. Marykripa Mathew

Ms. Senthil Vadivu

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1. Introduction

Agricultural dashboards are critical decision-making instruments for agricultural decision-makers. The dashboards provide a clear summary of the key performance indicators (KPIs), thereby enabling stakeholders to make effective decisions regarding production, resource utilization, and overall operation efficiency. The particular dashboard in question is designed to provide a clear and understandable perspective of agricultural production statistics, thereby enabling effective strategic planning and operation reconfigurations.



But, how do dashboards help us make decisions?

❖ **Giving a combined picture of key metrics:**

It combines information on production quantity, production area, productivity, and comparisons by country, enabling users to easily understand the overall situation of agricultural activities.

❖ **Enabling trend identification:**

By plotting data over time and across categories (countries, products), the dashboard facilitates the identification of trends and patterns that might not be obvious in raw data.



❖ **Facilitating comparative analysis:**

The dashboard facilitates direct comparison of country and product categories' performance, allowing users to recognize best practices and areas where they can improve

❖ **Facilitating supporting resource allocation:**

Through giving insights into area and productivity use, the dashboard can inform decisions regarding resource allocation, including land, labor, and capital

❖ **Supporting forecasting:**

The dashboard features a component for forecasting production in the future, which can support long-term planning and strategic decision-making

❖ **Facilitating enhancing situational awareness:**

The dashboard shows current or near-real-time agricultural activities, allowing stakeholders to react rapidly to variations in conditions.

The dashboard design with its compact presentation and easy-to-read organization makes decision-makers gain real-time access and insights to what they require instantaneously. This enables them to make educated data-driven decisions that can help drive agricultural productivity, sustainability, and profitability.

2. Objectives

The main purpose of this agriculture dashboard is to present a holistic and easily understandable view of major agricultural indicators to allow stakeholders to make informed production, resource utilization, and strategic planning decisions.

Target Audience:

- ❖ **Agricultural policymakers:** To monitor national and regional agricultural performance and formulate effective policies.
- ❖ **Farm managers and producers:** To optimize production processes, improve resource utilization, and enhance profitability.
- ❖ **Agricultural researchers and analysts:** To identify trends, patterns, and areas for further investigation.
- ❖ **Investors and financial institutions:** To assess the viability and potential of agricultural projects.
- ❖ **Supply chain managers:** To ensure efficient sourcing and distribution of agricultural products.

Specific Metrics and Data Points:

- ❖ **Country-wise Area of Production:** Displays the geographical distribution of agricultural production, highlighting major producing regions.
- ❖ **Product-wise Production:** Shows the breakdown of production by product category (fruits and vegetables), indicating the relative contribution of each category.
- ❖ **Country-wise Production:** Presents the total production volume for each country, enabling comparisons and identification of leading producers.
- ❖ **Sum of Productivity:** Measures the efficiency of agricultural production, indicating the output per unit of input.
- ❖ **Average Production Country-wise:** Provides an average production figure for each country, facilitating comparisons and identifying outliers.
- ❖ **Average Area Used Country-wise:** Shows the average land area used for production in each country, indicating resource utilization.
- ❖ **Country-wise Use of Area:** Displays the total land area used for agricultural production in each country.
- ❖ **Prediction of Country-wise Production:** Forecasts future production trends, aiding in long-term planning.
- ❖ **Total Area Used for Production:** Provides an overall measure of land utilization for agriculture.
- ❖ **Count of Countries:** Indicates the number of countries included in the dataset.

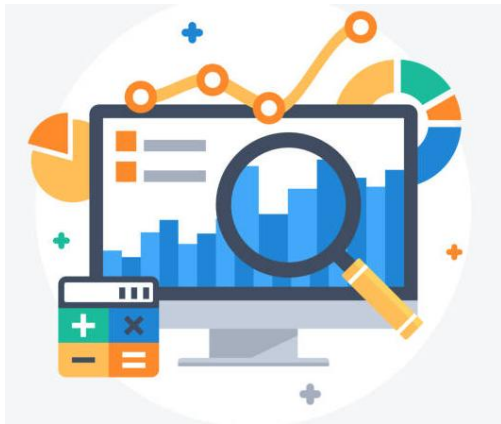


The dashboard aims to display these metrics in a visually appealing and easily understandable manner, using charts, graphs, and numerical summaries. This ensures that users can quickly grasp the key insights and make informed decisions.

3. Data Description

Data drawn to develop the agricultural dashboard comprise important measures dealing with agricultural output in different nations. It holds data on space utilized for manufacturing, production volumes, and attendant productivity for alternative products (fruit and vegetables) during the 2009-10 and 2010-11 years.

Key Data Elements:



- ❖ **Country:** The geographical location of agricultural production.
- ❖ **Product:** The type of agricultural product, categorized as either fruits or vegetables.
- ❖ **Year:** The period during which the data was collected (2009-10 and 2010-11).
- ❖ **Area (in Hectares):** The land area used for agricultural production.
- ❖ **Production (in Tonnes):** The total volume of agricultural output.
- ❖ **Productivity:** The efficiency of production, calculated as the output per unit of input (likely tonnes per hectare).

Data Characteristics:

- ❖ **Categorical Data:** Country and Product are categorical variables, enabling grouping and comparison between various regions and product types.
- ❖ **Numerical Data:** Area, Production, and Productivity are numerical variables, offering quantitative measures of agricultural performance.
- ❖ **Time-Series Data:** The data covers two years, allowing the assessment of trends and changes over time.
- ❖ **Missing Data:** There are some missing values in the data set, specifically in the production and productivity columns for some countries and years. This might need imputation or other data pre-processing methods to enable proper analysis.
- ❖ **Variability:** The data show high variability by country and product category, due to variations in agricultural practices, climate, and availability of resources.

Data Source and Collection:

The information was taken from data.gov.in, which is India's official open government data portal. The portal gives access to numerous datasets across various government departments and ministries. The dataset applied for this dashboard would probably have been gathered and disseminated by the Ministry of Agriculture & Farmers Welfare, Government of India. The information would have been gathered via agricultural surveys, censuses, or other official methods of gathering information.

Data Quality:

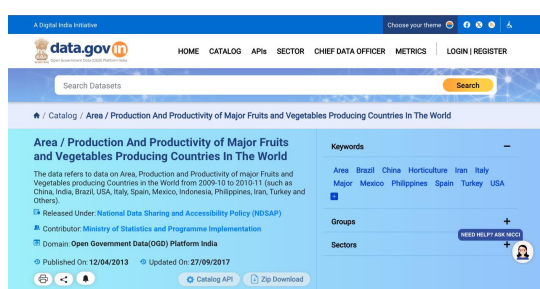
The quality of the data is crucial for the accuracy and reliability of the dashboard. The presence of missing values and potential inconsistencies should be addressed to ensure that the analysis and interpretations are valid.

4. Methodology with visuals

The creation of the agricultural dashboard involves several steps, from data collection and preprocessing to visualization and interpretation.

Step 1: Data Collection and Pre-processing

- ❖ **Data Gathering:** Collecting secondary data from data.gov.in, India's official open government data portal. The link for the data: [Agricultural Dataset](#). Download the Dataset and Rename it as “**Babitha Ravindra - 2441616 - Data Visualization - CIA 3 - Agricultural Dataset**”. Rename the first sheet as “Original Dataset”.



Country	Fruits 2009-10 Area (In Hectare)	Fruits 2010-11 Area (In Hectare)	Fruits2009-10 Production (In Tonnes)	Fruits2010-11 Production (In Tonnes)	Fruits2009-10 Productivity (In Tonnes/Hectare)	Fruits2010-11 Productivity (In Tonnes/Hectare)	Vegetables 2009-10 Area (In Hectare)	Vegetables 2010-11 Area (In Hectare)
China	11057621	11402073	109615629	122184944	9.9	10.7	24080078	21047200
India	6329200	6383000	71515500	74878000	11.3	11.7	7984800	8495000
Brazil	2449758	2441430	38683456	39286781	15.8	16.1	1158911	1123800
USA	1159589	1145475	27762483	25383917	23.9	22.2	1106103	1089805
Italy	1267386	1277347	17646110	16907895	13.9	13.2	592757	
Spain	1743165	1562160	16277697	15184420	9.3	9.699999999999999	523843	536799

- ❖ **Data Transformation:** It refers to converting data into a suitable format for analysis and visualization. This involves aggregating data, calculating derived metrics (e.g., productivity), or transforming data types. Open a new sheet in the workbook called “Transformed Dataset” and store all transformed details in that sheet.

	A	B	C	D	E	F
1	Country	Fruits 2009-10 Area (In Hectare)	Fruits 2010-11 Area (In Hectare)	Fruits2009-10 Production (In Tonnes)	Fruits2010-11 Production (In Tonnes)	Fruits2009-10 Productivity (In Tonnes/Hectare)
2	China	11057621	11402073	109615629	122184944	9.9
3	India	6329200	6383000	71515500	74878000	11.3
4	Brazil	2449758	2441430	38683456	39286781	15.8
5	USA	1159589	1145475	27762483	25383917	23.9
6	Italy	1267386	1277347	17646110	16907895	13.9
7	Spain	1743165	1562160	16277697	15184420	9.3
8	Mexico	1201897	1212387	16122211	15255598	13.4
9	Indonesia	721258	662800	16028273	14867762	22.2
10	Philippines	1123530	1155871	15620664	16181731	13.9
11	Iran	1264574	NA	13183703	NA	10.4
12	Turkey	1036528	1092623	12829520	13945604	12.4
13	Others	26242703	26751915	228947458	245223757	8.7
14	World+	55011784	55087081	579926404	599300409	10.5

A SECTION OF THE ORIGINAL DATASET

COUNTRY	PRODUCT	YEAR	AREA (IN HECTARES)	PRODUCTION (IN TONNES)	PRODUCTIVITY
China	FRUITS	2009-10	11057621	109615629	9.91
India	FRUITS	2009-10	6329200	71515500	11.30
Brazil	FRUITS	2009-10	2449758	38683456	15.79
USA	FRUITS	2009-10	1159589	27762483	23.94
Italy	FRUITS	2009-10	1267386	17646110	13.92
Spain	FRUITS	2009-10	1743165	16277697	9.34
Mexico	FRUITS	2009-10	1201897	16122211	13.41
Indonesia	FRUITS	2009-10	721258	16028273	22.22
Philippines	FRUITS	2009-10	1123530	15620664	13.90
Iran	FRUITS	2009-10	1264574	13183703	10.43
Turkey	FRUITS	2009-10	1036528	12829520	12.38
Others	FRUITS	2009-10	26242703	228947458	8.72
China	FRUITS	2010-11	11402073	122184944	10.72
India	FRUITS	2010-11	6383000	74878000	11.73
Brazil	FRUITS	2010-11	2441430	39286781	16.09

A SECTION OF THE TRANSFORMED DATASET

- ❖ **Data Cleaning:** It refers to addressing missing values, inconsistencies, and outliers. This involves imputation, removal of erroneous data, or other data cleaning techniques. For the corrective measure as the missing data in the dataset for the missing values in 2010 for fruits & vegetables and Iran and Turkey respectively have been entirely removed. Open a new sheet in the workbook called “Cleaned Dataset” and store all the changes in that sheet.

Step 2: Data Analysis

- ❖ **Comparative Analysis:** Compare agricultural performance across countries and product categories, identifying differences and similarities.
- ❖ **Trend Analysis:** Analyze changes in agricultural metrics over time, identifying patterns and trends.
- ❖ **Correlation Analysis:** Examine relationships between different variables, such as the correlation between area used and production volume.



Step 3: Dashboard Creation

- **Select Visualization Tools:** Choose appropriate software or tools for creating the dashboard for e.g., Tableau, Power BI, Excel. Here, we've chosen Power BI.
- **Create a New Report:** Open Power BI and choose to create a new report.
- **Import Data into Power BI:**



1. Open Power BI Desktop.
2. On the "Home" tab, click "Get Data."
3. Choose the data source (here, Excel).
4. Select the file with the name [“Babitha Ravindra - 2441616 - Data Visualization - CIA 3 - Agricultural Dataset”](#) and choose the “Transformed Dataset” sheet containing the data.
5. Click "Load."

- **Create Background Image:**

1. Open Microsoft PowerPoint.
2. On the “Designs” tab, click on all “Background Color” option and customize it using this hex code - “#4c956c”.
3. Then go to the “Insert” Tab and choose the “Shapes” option and click on the rectangle with curved edges option.
4. Add those rectangles according to planned layout and customize their colors to “#2c6e49”.
5. Now save the PPT in PNG format as “Dashboard Background”.

- **Add Background Image to the Dashboard:**

1. Go to the "View" tab.
2. Click on the "Format" icon.

3. In the "Page Background" section, locate "Wallpaper."
4. Click the "Add Image" button.
5. Browse your computer and select the "Dashboard Background" PNG image file.
6. Adjust the transparency of the image to 0 using the slider.
7. Choose how the image fits the canvas as "Fill" from the "Image fit" dropdown.

➤ Creating Visuals:

1. Title:

1. Go to the "Insert" tab and click "Text Box."
2. Type "AGRICULTURAL DASHBOARD."
3. Format the text (font, size, color) to match the image.
4. Position it at the top.



2. KPI Cards (0.02K, 68946.25K, 4778.42K):



1. Go to the "Visualizations" pane.
2. Click the "Card" visual.
3. Drag the relevant measure "Average Production" to the "Fields" area of the Card visual.
4. Format the card:

- **Callout Value:** Adjust font size, color, and display units (e.g., K for thousands).
- **Category Label:** Add "Average..." or a similar label.
- **Background:** Set a dark background.
- **Border:** Add a light border if needed.

5. Repeat for each KPI card.

3. Year Slicers (2009, 2010):

1. Go to the "Visualizations" pane.
2. Click the "Slicer" visual.
3. Drag the "Year" field to the "Field" area of the Slicer.
4. Format the slicer:



- **Slicer Settings:** Choose "Horizontal" orientation.
- **Selection Controls:** Enable "Single Select."
- **Items:** Adjust font size and color.
- **Background:** Set a dark background.
- **Border:** Add a light border if needed.

4. Country-Wise Area of Production Map

• Map Visual:

- Go to the "Visualizations" pane.
- Click the "Map" visual.



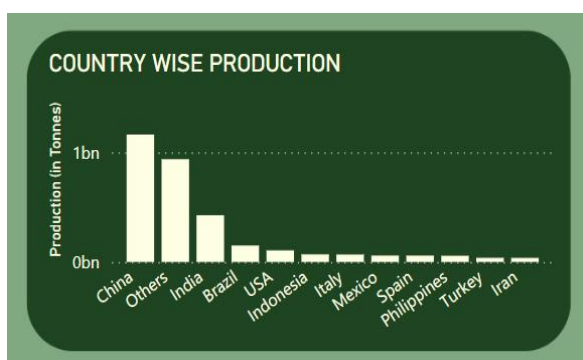
- Drag "Country" to the "Location" field.
- Drag "Area Used" to the "Size" field.
- Drag "Product" to the "Legend" field.
- **Formatting:**
 - **Map Styles:** Choose a dark map style.
 - **Data Colors:** Assign distinct colors to "Fruits" and "Vegetables".
 - **Category Labels:** Customize the labels to show product names.
 - **Bubble Sizes:** Adjust the size of the bubbles to represent area used.

5. Product-Wise Production Donut Chart

- **Donut Chart:**
 -
 - Go to the "Visualizations" pane.
 - Click the "Donut Chart" visual.
 - Drag "Product" to the "Legend" field.
 - Drag "Production Amount" to the "Values" field.
- **Formatting:**
 - **Data Colors:** Assign distinct colors to "Fruits" and "Vegetables."
 - **Detail Labels:** Show category and percentage.
 - **Title:** Add "PRODUCT WISE PRODUCTION."
 - **Background:** Set a dark background.
 - **Border:** Add a light border if needed.



6. Country-Wise Total Production Bar Chart



- **Bar Chart:**
 - Go to the "Visualizations" pane.
 - Click the "Clustered Bar Chart" visual.
 - Drag "Country" to the "Axis" field.
 - Drag "Production Amount" to the "Values" field.
- **Formatting:**
 - **Data Colors:** Customize the colors of the bars.
 - **Title:** Add "COUNTRY WISE PRODUCTION."
 - **Axis Labels:** Adjust font size and color.
 - **Data Labels:** Show the production values on the bars.
 - **Background:** Set a dark background.
 - **Border:** Add a light border if needed.

7. Sum of Productivity Table

- **Table Visual:**
 - Go to the "Visualizations" pane.
 - Click the "Table" visual.
 - Drag "Country" to the "Values" area.
 - Drag "Sum of Productivity" (create a measure if needed) to the "Values" area.
- **Formatting:**
 - **Grid:** Adjust the grid style and colors.
 - **Column Headers:** Adjust font size and color.
 - **Values:** Adjust font size and color.
 - **Background:** Set a dark background.
 - **Border:** Add a light border if needed.



COUNTRY	Sum of PRODUCTIVITY
Brazil	95.43
China	62.12
India	57.03
Indonesia	96.87
Iran	46.99
Italy	78.66
Mexico	62.12
Others	44.12
Philippines	100.90
Spain	70.35
Total	870.80

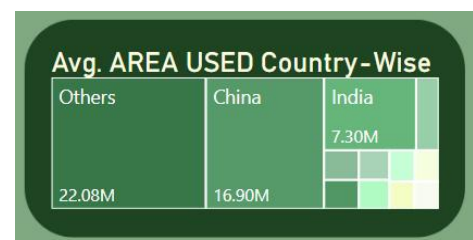
8. Average Production Country-Wise Bar Chart



- **Bar Chart:**
 - Go to the "Visualizations" pane.
 - Click the "Clustered Bar Chart" visual.
 - Drag "Country" to the "Axis" field.
 - Drag "Average Production" (create a measure if needed) to the "Values" field.
- **Formatting:**
 - **Data Colors:** Customize the colors of the bars.
 - **Title:** Add "Avg. PRODUCTION COUNTRY - WISE."
 - **Axis Labels:** Adjust font size and color.
 - **Data Labels:** Show the average production values on the bars.
 - **Background:** Set a dark background.
 - **Border:** Add a light border if needed.

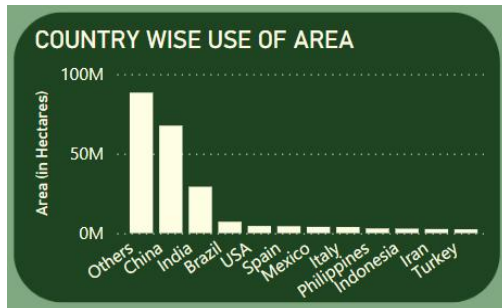
9. Average Area Used Country-Wise Bar Chart

- **Treemap Visual:**
 1. Go to the "Visualizations" pane.
 2. Click the "Treemap" visual.
 3. Drag "Country" to the "Group" field.
 4. Drag "Average Area Used" to the "Values" field.
- **Formatting:**
 - **Data Colors:** Customize the colors.
 - **Title:** Add "Avg. AREA USED Country-Wise."
 - **Labels:** Show the labels.



- **Background:** Set a dark background.
- **Border:** Add a light border if needed.

10. Country-Wise Use of Area Bar Chart



- **Bar Chart:**

- Go to the "Visualizations" pane.
- Click the "Clustered Bar Chart" visual.
- Drag "Country" to the "Axis" field.
- Drag "Aggregate Area" (create a measure if needed) to the "Values" field.

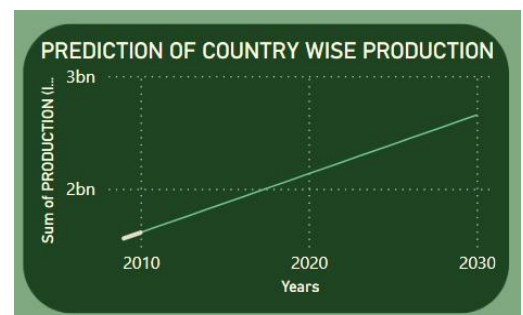
- **Formatting:**

- **Data Colors:** Customize the colors of the lines.
- **Title:** Add "COUNTRY WISE USE OF AREA."
- **Axis Labels:** Adjust font size and color.
- **Data Labels:** Show the area used values on the bars.
- **Background:** Set a dark background.
- **Border:** Add a light border if needed.

11. Prediction of Country Wise Production Line Chart

- **Line Chart:**

- Go to the "Visualizations" pane.
- Click the "Line Chart" visual.
- Drag "Year" to the "Axis" field.
- Drag "Predicted Production" (create a measure using DAX for forecasting) to the "Values" field.

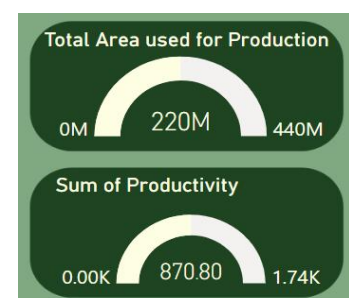


- **Formatting:**

- **Data Colors:** Customize the color of the line.
- **Title:** Add "PREDICTION OF COUNTRY WISE PRODUCTION."
- **Axis Labels:** Adjust font size and color.
- **Data Labels:** Show the predicted production values on the line.
- **Background:** Set a dark background.

12. Gauge Visuals (for "Average Productivity," "Average Production," and "Average Area Used"):

1. Go to the "Visualizations" pane.
2. Click the "Gauge" visual.
3. Drag "Total Area" to the "Value" field.
4. Repeat steps 2 and 3 for "Sum of Productivity"
5. Apply Same formatting



5. Interpretation

The key points we can focus from our data is:

- ❖ **Production Growth:** Overall, there seems to be a positive trend in production volume from 2009-10 to 2010-11, indicating growth in the agricultural sector.
- ❖ **Area vs. Productivity:** While some countries achieve high production through extensive land use, others demonstrate high productivity with less land, suggesting efficient resource utilization.
- ❖ **Regional Disparities:** Significant disparities exist in agricultural performance across different regions. Some regions show strong growth and high productivity, while others lag behind, indicating potential areas for development and investment.
- ❖ **Product Focus:** The relative contribution of fruits and vegetables to overall production varies across countries, reflecting differences in agricultural practices and consumer preferences.
- ❖ **Emerging Trends:** The prediction lines on the dashboard suggest that the dominance of China and India in agricultural production is likely to continue in the coming years.



According to the data and the dashboard the problems posing agriculture our:

- ❖ **Productivity Gap:** The high level of disparity in productivity between countries indicates that there exists a broad productivity gap between high-performance and low-performance countries. This means that there is potential for improvement in the majority of regions through the adoption of enhanced agriculture practices, technology, and resource management.
- ❖ **Land Utilization Efficiency:** While other countries achieve a high rate of agricultural production via extensive land use, it is not sustainable in the long term. Land utilization efficiency and sustainable agriculture should be improved to maintain food security and environmental integrity.
- ❖ **Regional Imbalances:** The variation in farm performance between regions suggests that interventions and investment should be directed towards the prevailing problems in the lagging regions. This could involve infrastructure development, capacity building, and access to technology and markets.
- ❖ **Climate Change Vulnerability:** The data itself isn't so much a declaration of the impact of climate change. Yet it would be worthy of mention that agriculture is most exposed to climate change, and it must develop adaptation and resilience in the face of the new realities. These could be in terms of encouraging climate-smart agriculture, cultivation of drought-resistant crops, and water management.
- ❖ **Data Gaps:** Gaps in data in the dataset highlight the need for improving data collection and reporting mechanisms. Accurate and detailed data is most critical for effective monitoring, evaluation, and decision-making for agriculture.



6. Suggestions and Forecasting

Suggestions for Minimizing Identified Problems:

- ❖ **Promote Sustainable Intensification:** Support agriculture that enhances productivity without increasing land area, for example, precision agriculture, integrated pest management, and conservation farming practices.
- ❖ **Invest in Research and Development:** Invest in research and development of new crop varieties, technologies, and management practices that increase productivity, resource use efficiency, and climate resilience.
- ❖ **Improving Infrastructure:** Improve rural infrastructure by building irrigation, storage facilities, and transportation systems to allow for market access and reduce post-harvest loss.
- ❖ **Empower Farmers:** Offer farmers access to information, training, and financial services to facilitate them in adopting improved agricultural practices and technologies.
- ❖ **Encourage Diversification:** Encourage diversification of farm production to minimize climate change and market risks.
- ❖ **Strengthen Data Gathering and Monitoring:** Strengthen data gathering and monitoring systems to furnish timely and correct information to aid decision-making.



Predictive Analysis (Based on the Dashboard Forecasts):

- ❖ **Sustained Primacy of India and China:** The prediction is that India and China will continue to be the world's dominant agricultural producers in the future. This highlights the importance of these countries in the global food security landscape.
- ❖ **Expansion in Production:** The forecast trends show an additional rise in farm production, but the growth rates can vary in various countries and products.
- ❖ **Space for New Emerging Producers:** Although China and India are likely to continue their dominance, new nations are likely prospects for emerging producers, especially nations with good agricultural conditions and rising investments in agriculture.
- ❖ **Uncertainty and Volatility:** It is important to note that forecasts tend to be prone to volatility and uncertainty because of the nature of agricultural produce being subject to a range of factors, including climate change, technological advancements, and market forces.



7. Conclusion

The agricultural dashboard is a useful resource for comprehension and analysis of agricultural production statistics. It discloses important trends, patterns, and issues in the industry, allowing stakeholders to make effective production, resource management, and strategic decisions. The analysis points to China and India's leadership in agricultural production, country-to-country variation in productivity, and the importance of enhancing land use efficiency and sustainable methods. The dashboard also forecasts sustained growth in farm production, albeit with uncertainties owing to climate change and other issues.

The information derived from the dashboard is complemented by the efficient visualization and data processing methods used. The application of multiple chart types (maps, bar charts, line charts, treemaps, bubble charts) enables effective and concise communication of intricate agricultural information, making it easy to compare and spot trends. Interactive elements such as filters and tooltips also increase user interaction and discovery. Additionally, the meticulous data cleaning and preprocessing steps, including handling missing values and computing productivity, guarantee the validity and dependability of the conclusions drawn from the dashboard. The link to the final Dashboard: [Agricultural Dashboard](#)

These findings, together with the successful visualization and data analysis methods, can guide policy actions, investment strategies, and research agendas to confront the challenges and opportunities facing the agricultural sector. Through the support of sustainable intensification, investment in research and development, enhanced infrastructure, and empowerment of farmers, we can guarantee food security and improve the livelihoods of millions of individuals globally.

In short, by ongoing observation and study of agricultural statistics, we may better comprehend the intricate dynamics of the sector and make well-informed decisions to promote sustainable and resilient farming systems.

RESULTANT DASHBOARD:



DASHBOARD WITH FILTER (Year: 2009)**DASHBOARD WITH FILTER (Year: 2010)**